

THE GREEDY 3-SUMFREE META-CONJECTURE: A PROOF, SHARP ON THE RANGE

$$g \geq d$$

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ABSTRACT. Bosma, Bruin, Fokkink, Grube, Reuijl and Tromp [2] conjecture (Conjecture 17, their “meta-conjecture”) a closed periodic description of the greedy 3-sumfree sequence $S_{1,g,g+d}$ for every $d \geq 2$ and $g \geq d+1$, and report that they could not verify it uniformly in (g, d) because their decision procedure, Walnut, rejects the required term $k \cdot (5g+2d)$ as a product of two variables. We prove the conjecture, and show it remains true down to $g = d$: writing $G(g, d) = \{1, g\} \cup [g+d, 2g+d] \cup \bigcup_{k \geq 1} (k(5g+2d) + [g+d-2, 2g+d-2])$ for the conjectured answer set, we prove $G(g, d) = S_{1,g,g+d}$ iff $d \geq 2$, $g \geq d$ and $(g, d) \neq (2, 2)$ — exactly the range on which it can hold. This is strictly larger than the conjectured hypothesis $g \geq d+1$: it adds the diagonal $g = d$, $d \geq 3$. The proof is by hand, an interval-sumset argument in the style O. Shtrezi used for the $d = 1$ companion conjecture, and independently by machine: the paper’s obstruction is removed by a one-line change of variables that turns the parametrised-modulus statement into ordinary Presburger arithmetic, decided by the Peanut decision procedure as eight sentences quantified over all (g, d) at once (6.2s total), and reproduced in Walnut 8-dev. Every finite claim was independently re-verified, by a second implementation sharing no code with the first, over thousands of (g, d) cells and by a second decision procedure.

1. BACKGROUND AND THE CONJECTURE

For integers $x < y < z$, the *greedy 3-sumfree sequence* $S_{x,y,z}$ is the increasing sequence beginning x, y, z in which each later term is the least integer exceeding the previous term that is *not* a sum of three distinct earlier terms. $S_{1,2,3}$ and $S_{1,3,4}$ are OEIS [A026471](#) and [A026475](#). Bosma, Bruin, Fokkink, Grube, Reuijl and Tromp [2] study these sequences with the automatic-sequence decision procedure Walnut, proving closed forms for $S_{1,2,3}$, $S_{1,3,4}$, $S_{1,4,5}$ (their Theorem 15) and, for the family $S_{1,g,g+1}$, a conjecture (their Conjecture 16) later proved by O. Shtrezi [5]. For the two-parameter family $S_{1,g,g+d}$, $d \geq 2$, they report “firm computational evidence, from an implementation in Magma”, for the following, quoted verbatim:

Conjecture 1.1 ([2, Conjecture 17]). *Let $d \geq 2$. For every $g \geq d+1$ the greedy 3-sumfree sequence $S_{1,g,g+d}$ is characterized as follows:*

$$z \in S_{1,g,g+d} \iff z \in \{1, g, 2g+d-1, 2g+d\} \text{ or } z \geq g+d \text{ and } z \bmod 5g+2d \in \{g+d-2, g+d-1, \dots, 2g+d-2\}.$$

In particular, for $d \geq 2$ and every $g \geq d+1$ after the first $g+3$ entries (in a preperiod) the sequence $S_{1,g,g+d}$ modulo $5g+2d$ is periodic with period $g+1$.

(The arXiv preprint version 1 [2] prints $z > g+d$ where the published Journal of Integer Sequences text has $z \geq g+d$; this is a typo in v1, since $z = g+d$ is a seed of the sequence and so must be admitted; the paper’s own worked example uses $z \geq 8 = g+d$. We use the published $\geq$ form throughout, as does the statement above.) The authors state that Walnut cannot verify Conjecture 1.1 uniformly in (g, d) : the term $z = k \cdot (5g+2d) + w$ needs the operator $*$ applied to two variables (k and g , or k and d), which their tool rejects outright, though “whenever a numerical value is substituted for $g \dots$ all is fine”; they verify individual instances this way. As of August 2026 no treatment of Conjecture 1.1 for general (g, d) was found in the literature; the only citing work, Shtrezi’s proof of the $d = 1$ case [5], does not mention $d \geq 2$ or the meta-conjecture, and the paper’s own Remarks 18 says the $d = 1$ family is *not* a special case of Conjecture 1.1. The conjecture was open.

2. THE THEOREM

For $d \geq 2$, $g \geq 2$ write $m = 5g+2d$ and let $G(g, d)$ be the right-hand side of Conjecture 1.1, viz.

$$G(g, d) = \{1, g\} \cup [g+d, 2g+d] \cup \bigcup_{k \geq 1} (km + [g+d-2, 2g+d-2]).$$

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Theorem 2.1. *Let $d \geq 2$ and $g \geq 2$. Then $G(g, d) = S_{1,g,g+d}$ if and only if $g \geq d$ and $2g \geq d + 3$; equivalently, if and only if $g \geq d$ and $(g, d) \neq (2, 2)$.*

In one sentence: Conjecture 1.1 is true exactly as stated, for $d \geq 2$, $g \geq d + 1$, and Theorem 2.1 shows it remains true on the strictly larger range $d \geq 2$, $g \geq d$, $(g, d) \neq (2, 2)$ — which is exactly the range on which the description *can* be correct, since Theorem 2.1 is an “if and only if”. The extra range is the diagonal $g = d$, $d \geq 3$ (e.g. $S_{1,3,6}$, $S_{1,4,8}$, $S_{1,5,10}$, $\dots$), which Conjecture 1.1’s hypothesis $g \geq d + 1$ excludes; amusingly the paper’s own worked verification, following its Example 19, is the instance $(g, d) = (4, 4)$, on that diagonal and outside the conjecture’s own hypothesis.

Corollary 2.2. *For $d \geq 2$, $g \geq d$, $(g, d) \neq (2, 2)$, $S_{1,g,g+d}$ begins with the $g+3$ terms $1, g, g+d, g+d+1, \dots, 2g+d$ and from there repeats, modulo $m = 5g + 2d$, the block of $g+1$ residues $g+d-2, g+d-1, \dots, 2g+d-2$, one block per period — the explicit form of Conjecture 1.1’s “periodic with period $g+1$ ”.*

3. PROOF

The obstruction, and its removal. Fixing g, d numerically, $z = k \cdot m + w$ is an ordinary linear (Presburger) term and individual instances of Conjecture 1.1 verify at once, as the source paper does. With g, d free, $k \cdot (5g + 2d)$ is a product of variables and is rejected outright. The fix needs no new mathematics, only a change of variables: never name z , but carry every integer as a (quotient, remainder) pair against m .

Lemma 3.1 (Carry split). *Let $m \geq 1$ and let $a = k_a m + w_a$, $b = k_b m + w_b$, $c = k_c m + w_c$, $z = k_z m + w_z$ with $0 \leq w_a, w_b, w_c, w_z < m$. Then*

$$a + b + c = z \iff \exists j \in \{0, 1, 2\} : w_a + w_b + w_c = w_z + jm \text{ and } k_a + k_b + k_c + j = k_z.$$

Proof. Put $j = k_z - (k_a + k_b + k_c)$. If $a + b + c = z$ then $jm = (w_a + w_b + w_c) - w_z$, and the right side lies in $[-(m-1), 3m-3]$, forcing $-1 < j < 3$, i.e. $j \in \{0, 1, 2\}$; this j is unique since jm determines it, and the displayed equalities hold. The converse is immediate. $\square$

Since j ranges over the fixed finite set $\{0, 1, 2\}$, each disjunct’s $jm = j(5g + 2d)$ is *linear* in g, d : the product of variables never reappears. This is folklore — the schoolbook carry argument for sums of a bounded number r of terms, giving $j \in \{0, \dots, r-1\}$ in general — and is not claimed as new mathematics; the contribution is recognising that it removes exactly the blocker [2] reports. It is, as the underlying attack document puts it, “a reformulation, not a prover capability”.

The pen-and-paper route. Write $B = [g + d, 2g + d]$ ($g + 1$ integers) and $A_k = km + [g + d - 2, 2g + d - 2]$ for $k \geq 1$, so $G(g, d) = \{1, g\} \cup B \cup \bigcup_{k \geq 1} A_k$. Using $2^X = \{x + y : x, y \in X, x < y\}$ and $3^X = \{x + y + z : x, y, z \in X, x < y < z\}$ (as in [5]), the complement of G above the seeds splits into gaps $\text{Gap}_0 = [2g + d + 1, 6g + 3d - 3]$ and $\text{Gap}_k = km + [2g + d - 1, 6g + 3d - 3]$ for $k \geq 1$. Four propositions, each an interval computation, give Theorem 2.1 by induction:

Proposition 3.2 (Maximality). *If $d \geq 2$, $g \geq d$, $2g \geq d + 3$, every $z > g + d$ with $z \notin G$ is a sum of three distinct elements of G .*

Proposition 3.3 (3-sumfreeness). *If $d \geq 2$, $g \geq 2$, no element of G is a sum of three distinct elements of G . (Needs only $g, d \geq 2$, not $g \geq d$.)*

Proposition 3.4 (Seeds). *If $d \geq 2$, $g \geq 2$, then $G \cap [1, g + d] = \{1, g, g + d\}$.*

Proposition 3.5 (Sharpness). *If $2 \leq g < d$, then $z = 3g + d + 2$ satisfies $z > g + d$, $z \notin G$, and z is not a sum of three distinct elements of G ; hence $G \neq S_{1,g,g+d}$. The same holds at $(g, d) = (2, 2)$, at $z = 14$.*

Proposition 3.3 is proved by a twelve-case check: G splits into the four parts $\{1\}, \{g\}, B, A_k$, and every sum of three (not necessarily distinct-part) summands is shown to land inside a Gap, using $2^B = [2g + 2d + 1, 4g + 2d - 1]$, $3^B = [3g + 3d + 3, 6g + 3d - 3]$ and the analogous identities for the A_k ; the only place the hypothesis $d \geq 2$ is used is the lower end of the $1 + A_i + A_j$ case, which is exactly why $d = 1$ needs a different modulus (Shtrezi’s). Proposition 3.2 covers each $\text{Gap}_0/\text{Gap}_k$ by a chain of four sumset intervals whose junctions are exactly $g \geq d$ and $2g \geq d + 3$. Proposition 3.5’s witness $z = 3g + d + 2$ falls in Gap_0 but below every covering interval that needs $g < d$; the step from “ z is not a 3-sum” to “the greedy rule admits it, so $G \neq S$ ” additionally needs G and S to agree on $[1, z]$, which they do: the only part of Gap_0 below z is $[2g + d + 1, 3g + d + 1] = 1 + g + B$, covered with *no* hypothesis at all, so the induction below runs unobstructed up to z for every $g, d \geq 2$. Full case-by-case verification of all twelve rows and both covering chains is in [1, §4].

The four propositions combine by strong induction: Proposition 3.4 settles $[1, g + d]$; for $z > g + d$, every representation $z = a + b + c$ with $a, b, c \in G$ has $a, b, c < z$, so by the induction hypothesis they are exactly the earlier terms of $S_{1,g,g+d}$; then Proposition 3.2 (if $z \notin G$) or Proposition 3.3 (if $z \in G$) tells the greedy rule what to do, matching G . This proves Theorem 2.1’s “if”; the “only if” is Proposition 3.5 together with the trichotomy $g < d$, $(g, d) = (2, 2)$, or $g \geq d \wedge 2g \geq d + 3$, which is exhaustive for $d, g \geq 2$.

The machine route. The same statement is also settled by decision procedure, with no hand argument. Encode $k \cdot m + w \in G(g, d)$ as a predicate $\text{ING}(k, w)$ in (k, w, g, d) directly from the definition of G ; every disjunct forces $w \leq 2g + d < m$, so the encoding is injective (checked as obligation T1 below), which is what licenses Lemma 3.1 without an explicit $w < m$ guard on the summands. The sum and distinctness conditions become, via Lemma 3.1, ordinary linear Presburger formulas in ten free variables $(g, d, k_a, k_b, k_c, k_z, w_a, w_b, w_c, w_z)$. Propositions 3.2–3.5 (with a restricted, sound existential witness family for Proposition 3.2’s “there exist three distinct elements summing to z ”, since the fully unrestricted six-variable existential exhausts a 6 GB budget in both provers after ~ 280 s and remains out of reach) become eight closed first-order sentences T0–T7, each discharged by automaton compilation in the Peanut decision procedure in 6.2 s total (peak 1.3 GB), and independently reproduced — posed as universally quantified sentences, which forces the full ten-fold existential projection Peanut’s ‘witness’-on-open-formula route does not need — in Walnut 8-dev (52.7 s total). The full transcript, formulas, and driver are `explore/attack4_engine.py`, `explore/attack4_walnut.py`, `results/attack4_transcript.txt`, `results/attack4_engine.json` and `results/attack4_walnut_uniform.json`, all in the repository below; the full account is [1], §5.

4. VERIFICATION

Every claim above was re-derived and independently re-checked, with code sharing nothing with the implementation that produced it, in a referee pass whose transcript is [6]. The hand proof of §3 was re-derived step by step, including all twenty-four interval-endpoint inequalities and both covering chains, and no gap was found; the primary source [2] was independently re-fetched from the published J. Integer Seq. LaTeX source, not quoted from memory, and Conjecture 1.1’s wording and the $\geq$ -vs- $>$ point confirmed against it directly.

Check (independent implementation)	Scope	Result
Theorem 2.1’s iff vs. real greedy sequences	$d = 2 \dots 30$, $g = 2 \dots 200$, 40 periods of $5g + 2d$ (5771 cells)	0 mismatches
Proposition 3.5’s first-failure point $3g + d + 2$ (and 14 at $(2, 2)$)	every failing cell above	exact in all of them
adversarial spot checks, e.g. $(100, 100)$, $(200, 3)$, $(3, 200)$, $(52, 50)$	11 cells	0 mismatches
Propositions 3.2–3.5, Lemma 3.1, encoding injectivity, all twelve interval containments, both covering chains, restricted-witness soundness and completeness	$d = 2 \dots 14$, $g = 2 \dots 30$ (377 cells)	0 failures
Proposition 3.3 in the failing region $g < d$	same grid	never fails
$d = 1$ control, meta-conjecture’s own modulus $5g + 2$	$g = 2 \dots 11$	description <i>and</i> 3-sumfreeness both fail — confirms $d \geq 2$ is load-bearing

The eight obligations of §3 were also re-posed independently, as closed universally quantified sentences in a different (equivalent) predicate shape, and re-run in both Peanut and Walnut 8-dev: all TRUE, including the ten-variable 3-sumfreeness sentence (Peanut, 3.50 s, peak 1252 MB; Walnut, 41.8 s) and, transcribed verbatim from the published LaTeX source, the paper’s own worked $(g, d) = (4, 4)$ Walnut fragment (TRUE, 0.3 s) — two independent decision-procedure implementations, two independent formula shapes, agreeing on every check. The unrestricted six-variable existential for Proposition 3.2 was attempted again independently and killed by a memory watchdog at 4874 MB after 274.5 s, confirming it is genuinely out of reach with both provers as currently implemented, not an artefact of one encoding.

5. REMARKS

What is new, and what is not. §3’s pen-and-paper argument is a competent transposition of Shtrezi’s interval-sumset technique [5] to a different modulus and range, not a new technique; the partition of G , the gap structure, and the two-proposition induction are his. What is new is doing this for $d \geq 2$, which nobody had done, and the *exact* range: Proposition 3.5, which has no counterpart in [5], is what turns an implication into the sharp equivalence of Theorem 2.1 and extends Conjecture 1.1 to the diagonal $g = d$, $d \geq 3$. Lemma 3.1 is schoolbook mathematics;

its role here is a change of variables, not a theorem, and it should be credited as such — the honest place to put the credit is the reformulation, not any capability of the decision procedure. The explicit preperiod/period of Corollary 2.2, which [2] states only as a count, is a small genuine addition.

What is not settled. The *irregular* region $2 \leq g < d$ and $(g, d) = (2, 2)$ is left open: Theorem 2.1 shows the regular description $G(g, d)$ fails there, but says nothing about the correct description, which [2, Remarks 18] already notes is irregular (their Example 19, $S_{1,5,12}$, is periodic with modulus 321 and period 32 — not of the form $5g + 2d$). Start values other than 1 are untouched. Whether a better decision-procedure ladder can project the fully unrestricted six-variable existential of Proposition 3.2 directly, rather than via the restricted-but-sound witness family used here, remains open; §4 confirms it is out of reach in both Peanut and Walnut 8-dev as they currently stand.

Acknowledgement and declaration of AI usage. This note, and the work behind it, were produced with substantial use of a large language model. This note carries this declaration from its first version. Precisely: the models were Claude (Anthropic): **claude-fable-5** as the orchestrating model, with **claude-opus-5** and **claude-sonnet-5** sub-agents for search, implementation, refereeing and drafting, all used through the Claude Code agent framework; they were used for (i) the literature and primary-source search, including fetching and checking the published Journal of Integer Sequences LaTeX source directly rather than quoting it from memory; (ii) the design and implementation of Peanut, the open-source decision procedure with which every machine computation here was run; (iii) the discovery of the carry-split reformulation of Lemma 3.1, and the statement and proof of Theorem 2.1 and Propositions 3.2–3.5, following the interval-sumset technique of [5]; (iv) an adversarial “referee” pass, also carried out by the model, which re-derived every step by hand, re-read the primary source independently, wrote brute-force and decision-procedure checks sharing no code with the original implementation, and found and corrected several defects in an earlier draft of the underlying analysis; and (v) the drafting of this text, most of which is the model’s output lightly edited by the author. The author directed the work, chose the problem, read and takes responsibility for every statement, and ran the computations. All finite claims were verified by the independent implementations described in Section 4; the referee transcript is **research/referee-verdicts/attack4-verdict.md** in the repository below. Any errors are the author’s.

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